Expanding & Simplifying Single Brackets

Exam code: 0607

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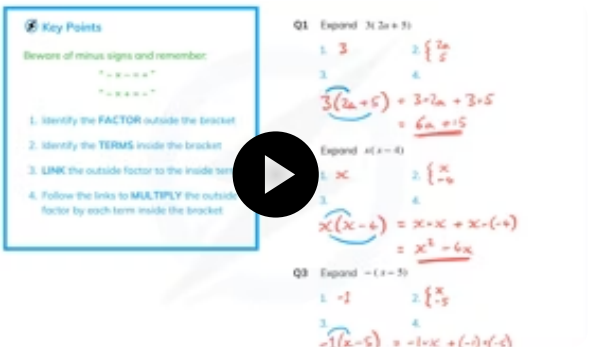

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Expanding one bracket

How do I expand a bracket?

Help me understand

- The expression $3x(x+2)$ means $3x$ **multiplied by** the bracket $(x+2)$
  - $3x$ is the term **outside** the bracket
    - this is sometimes called a **factor**
  - and $x+2$ are the terms **inside** the bracket
- Expanding the brackets means **multiplying** the **outside term** by **each term** on the **inside**
  - This will **remove** (get rid of) the **brackets**
  - $3x(x+2)$ expands to $3x \times x + 3x \times 2$ which simplifies to $3x^2 + 6x$
- Beware of **minus signs**
  - Remember the rules
    $- \times - = +$
    $- \times + = -$
  - It helps to put **brackets** around **negative** terms

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Answer:

Multiply the $4x$ term outside the brackets by both terms inside the brackets

$$4x \times 2x + 4x \times (-3)$$

Simplify

$$8x^2 - 12x$$

(b) Expand $-7x(4-5y)$.

Answer:

Multiply the $-7x$ outside the brackets by both terms inside the brackets

$$(-7x) \times 4 + (-7x) \times (-5y)$$

Simplify and remember that multiplying two negatives gives a positive

$$-28x + 35xy$$

Expand & simplify

How do I simplify brackets that are added together?

- First **expand** both brackets **separately**
  - $4(x+7) + 5x(3-x)$
    - The **first** set of brackets expands to $4 \times x + 4 \times 7$ which **simplifies** to $4x + 28$
    - The **second** set of brackets expands to $5x \times 3 + 5x \times (-x)$ which **simplifies** to $15x - 5x^2$
    - So $4(x+7) + 5x(3-x) = 4x + 28 + 15x - 5x^2$
- Then **collect like terms**
  - $4x + 15x = 19x$
    - The other two terms are **not** like terms
  - So $4(x+7) + 5x(3-x) = 19x + 28 - 5x^2$

Worked Example

(a) Expand and simplify $2(x+5) + 3x(x-8)$.

Answer:

Expand each set of brackets separately

You can keep negative terms inside brackets

$$2 \times x + 2 \times 5 + 3x \times x + 3x \times (-8)$$

Simplify each term

$$-22x + 10 + 3x^2$$

(b) Expand and simplify $3x(x+2) - 7(x-6)$.

Answer:

Expand each set of brackets separately
Be careful: the second set of brackets has a -7 in front, not +7

$$3x \times x + 3x \times 2 + (-7) \times x + (-7) \times (-6)$$

Simplify each term
Remember that multiplying two negatives gives a positive

$$3x^2 + 6x - 7x + 42$$

Collect like terms

$$3x^2 - x + 42$$

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